UIC Department of Chemical Engineering
UNIVERSITY OF ILLINOIS AT CHICAGO
COLLEGE OF ENGINEERING

TEAM ASSIGNMENT #2

Course: CHE 433. PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2019
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

Your name:____

Due Date: Friday September 20 by 5:00pm in my office

Number of points.............................100
Your score..................................

Note: **All problems are worth 100 points. You final score will be the average value**

Problem 1: Process Flowsheet Simulation Using Excel.

The production of cyclohexane is carried out via the hydrogenation of benzene in a bubble-column reactor in the presence of a nickel catalyst. The heat of the reaction is primarily removed by vaporization of the product. The remaining heat is removed by an external loop where the reactor reflux passes through a heat exchanger against boiling feed water (BFW) for the production of low pressure steam. Small amounts of methane are also formed from cracking of the benzene ring under the reaction conditions. The reaction mechanism can be for simplicity be represented by the following two reactions:

$$C_6H_6 + 3H_2 \rightarrow C_6H_{12}$$
$$C_6H_6 + 9H_2 \rightarrow 6CH_4$$

The conversion per pass of benzene is very high (99.5%) with very high per pass selectivity of 99.7%.We want to design a world class manufacturing plant for the production of high purity cyclohexane. High purity benzene feed is available from a nearby aromatics complex at $\left\{T = 100^o F \text{ and } P = 15\, psia\right\}$. The flow rate of benzene is taken as $F = 100\, lbmol/hr$ for our preliminary calculations. Hydrogen is available from a PSA unit at $\left\{T = 120^o F \text{ and } P = 335\, psia\right\}$ The flow rate of hydrogen is set at $F = 313\, lbmol/hr$ for now. The hydrogen purity is 97.5% with methane at 2.0% and nitrogen at 0.5%. The reactor operates at $\left\{T = 400^o F \text{ and } P = 315\, psia\right\}$. The reactor effluent is cooled at $\left\{T = 120^o F\right\}$ and flashed at high pressure $\left\{P = 310\, psia\right\}$. Hydrogen rich gas is recycled back to the reactor to maintain the hydrogen to benzene molar ratio at 3.3. A liquid stream is also recycled back to the reactor to convert the unreacted benzene. The liquid recycle stream flow rate cannot be more than about 45% of the fresh liquid benzene flow rate. The rest of the liquid is flashed at low pressure $\left\{P = 15\, psia\right\}$ through a throttling valve to produce pure cyclohexane and fuel gas. Complete the following tasks:

1. Using the **CHE433 Calculator** to prepare a material balance of the process. Clearly state all your operating data and the logic that you use to obtain flowsheet convergence
2. Estimate the overall conversion and selectivity
3. Estimate the mass flow rate of the final product and its purity
4. Comment whether the purity meets market specifications
5. Give an estimate of the Fixed Capital Investment (FCI) for a 200KMTA world class manufacturing plant in 2019 dollars.
6. Propose a method to remove the liquid recycle and redo the weight balance of the unit for the 200KMTA production. Calculate the reduction in the cost the raw material.

Thermodynamic Data

Component	High Pressure K-Values	Low Pressure K-values
Hydrogen	114.379	2464.97
Nitrogen	41.0817	840.15
Methane	15.5852	305.311
Benzene	0.0220743	0.315599
Cyclohexane	0.020994	0.295541

Problem 2: Getting Started with ASPEN Plus.

A fluid is flowing in a linear($L = 36\ ft$) $Schedule$ 40, 1″ standard pipe with a dirt factor of $k = 0.0077\,mm$. The Reynolds number is $\mathrm{Re} = 318{,}000$. Create an ASPEN Plus simulation to calculate the pressure drop along the linear pipe if the fluid is air that enters the pipe at $\left\{P_1 = 20\ psig, T_1 = 70\ ^oF\right\}$ and leaves the pipe at $T_2 = 170\ ^oF$. What is the rate of heat transfer to the air. Prepare a table and show the error in the calculation of the heat transfer and pressure drop, when compared with your hand calculation. Add an additional stream in your simulation that will represent the steam required for heating the air. Calculate the operating data for the steam line, i.e. flow rate, temperature and pressure. Verify the ASPEN Plus results for the steam line using hand calculations.